

Foundation Models for Classification of Soft Tissue Tumors

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Abstract. This research evaluates the characteristics of the SAM-Med3D and TabPFN as base models for small sample size classification of tumors on six clinical data sets (CRLM, desmoids, GIST, liposarcomas, liver, melanomas) with a focus on soft-tissue tumors. Volumetric scans from CT/MRI were resized to 128x128x128 by being cropped based on region-of-interest (ROI)-centric cropping. Three-dimensional encoder features were extracted using SAM-Med3D and compared with different pooling strategies using percentile pooling as the default method to produce 1920-dimensional case embeddings followed by standardization and principal component analysis. Classification was subsequently performed downstream with TabPFN and LoCalPFN, allowing in-context inference without requiring end-to-end re-training. Experiments evaluated the SAM-Med3D pipeline against two three-dimensional baselines (3D DenseNet-121 and 3D ViT), including an evaluation of attaching a light-weight classification head to a frozen SAM-Med3D encoder to evaluate feature quality directly. The results highlight current limitations of SAM-Med3D as a feature-extractor for tumor classification in clinically relevant work-flow.

Keywords: Foundation models · SAM-Med3D · TabPFN · Soft-tissue tumors

1 Introduction

Soft tissue tumors represent a heterogeneous group of lesions whose development, along with diagnosis and treatment, is characterized by significant uncertainty [14]. A major challenge in making accurate decisions about treatment and prognosis is obtaining a definitive, non-invasive diagnosis; however, many tumors exhibit common radiographic characteristics, thus requiring invasive techniques to confirm diagnosis, which may not be feasible prior to initiating therapy. Advances in medical imaging, coupled with advances in machine learning, have led to improved diagnostic capabilities; however, there are three primary challenges to applying this technology in a clinical setting: (1) lack of sufficient training data, (2) high processing demands for processing 3D medical image volumes

and (3) the requirement for universally applicable features that are stable [10, 2]. Training foundation models on extensive, diverse datasets of medical volumes may provide a means to address these issues since they generate adaptable embeddings that require less labeling information than traditional approaches, while reducing both training and inference costs and improving performance between different healthcare systems [17, 2].

In this work a large-scale, pretrained volumetric representation model such as SAM-Med3D, paired with compact, data-efficient classifiers, more specifically TabPFN and LoCalPFN are considered. The system re-uses semantic features generated by a foundation model for 3D medical imaging to minimize the quantity of task-specific labeled data needed, while a lesion-centric preprocessing technique, and a region-of-interest pooling technique focus the computational burden on the most informative aspects of the scan. The extracted embeddings are standardized, and potentially reduced in dimensionality. These standardized embeddings are then passed through transformer-based tabular classifiers [5] [16], that are specifically designed for the rapid testing of models using small, structured sets of features.

SAM-Med3D [17] was utilized in the proposed system as the feature extractor to generate rich embeddings from the regions of interest that contain the lesion, ROI pooling was applied with lesion filtering to produce compact feature vectors per study, and TabPFN [5] and LoCalPFN [16] were used as the classifiers. The system also includes complementary experiments comparing the use of transfer learning work flows, as opposed to utilizing native 3D classification heads trained on cropped volumes, allowing for an assessment of the trade-off between the reuse of existing features versus the utilization of end-to-end learning under matched preprocessing conditions.

The experimental data utilized consisted of 930 CT and MRI studies from the publically available WORC dataset [15], and included examples of several types of tumors including gastrointestinal stromal tumors, lipomatous tumors, desmoid-type fibromatosis, liver lesions, and melanoma metastases.

This research evaluates whether the use of pre-trained 3D embeddings in conjunction with transformer-based tabular classifiers, and limited quantities of labeled data, can achieve competitive levels of performance on medical classification tasks.[12, 10].

2 Data

2.1 The WORC Database

The imaging data used in this study originates from the WORC database, a publicly available collection of CT and MR scans with corresponding segmentations and clinical labels assembled specifically for radiomics research on soft tissue tumors [15]. The database comprises 930 patients from six independent radiomics studies conducted at the Erasmus MC. Three of these studies are about soft-tissue tumors, making it the largest publicly available resource for

soft tissue tumor classification research. Each case includes volumetric imaging data in NIfTI format, expert-delineated binary segmentation masks identifying one or more lesion regions, and clinical labels encoding diagnostic outcomes or tumor subtypes.

The database encompasses six distinct tumor categories, each representing a clinically relevant classification task. However, only three of the six cohorts are soft-tissue-tumor (STT) datasets: Lipomatous tumors (Lipo) (115 cases), Desmoid-type fibromatosis (203 cases), and Gastrointestinal stromal tumors (GIST) (246 cases).

The GIST cohort (246 cases) focuses on differentiating gastrointestinal stromal tumors from intra-abdominal gastrointestinal tumors that radiologically resemble GIST. The Lipomatous tumors cohort (115 cases) aims to differentiate benign lipomas from well-differentiated liposarcomas. The Desmoid-type fibromatosis cohort (203 cases) focuses on differentiating desmoid-type fibromatosis from extremity soft-tissue sarcomas.

The remaining three cohorts are oncologic lesion classification datasets included in WORC but are not STTs. The Liver cohort comprises 186 patients with primary solid liver tumors and addresses differentiation between malignant and benign lesions (malignant: hepatocellular carcinoma or intrahepatic cholangiocarcinoma; benign: hepatocellular adenoma or focal nodular hyperplasia). Colorectal liver metastases (CRLM) comprise 77 patients (93 lesions) with labels corresponding to histopathological growth patterns (100% desmoplastic versus 100% replacement). The Melanoma cohort comprises 103 patients with lung metastases of metastatic melanoma, with labels indicating BRAF mutation status.

2.2 Why WORC: A Challenging Benchmark

The WORC database served as the research choice because it presents one of the most difficult medical image classification benchmarks which researchers can access freely. The dataset contains specific features which make it challenging to work with, but more importantly, it is closer to real-world clinical tasks where foundation models are not adequately evaluated, thus making it an appropriate environment to assess foundation model performance in realistic medical applications.

The classification tasks involve subtle imaging differences that are difficult even for expert radiologists to distinguish without histopathological confirmation. The process of identifying benign lipomas from well-differentiated liposarcomas demands pathologists to identify small details which appear identical between these two tumor types [9]. The evaluation of GIST tumor mutation status through imaging data represents the maximum extent which visual characteristics can disclose about molecular tumor properties [7]. If foundation models can achieve reasonable performance on these ambiguous tasks, they are likely to generalize to a wide range of clinical applications where the imaging signal is more pronounced.

The database shows wide variation in imaging methods and scanner brands and scanning methods which affect the six different tumor types. The different characteristics in this data set represent what clinicians experience during their work because they cannot follow standardized procedures which creates difficulties for machine learning models to learn from protocol-based data. A model that achieves high performance in all six categories of WORC demonstrates the stability required for real-world deployment.

The study contains small tumor category sample sizes, including between 77 and 246 cases for each category. While traditional deep learning approaches require large datasets, pre-trained foundation models are particularly well-suited for this data-scarce scenario, as they leverage knowledge from large-scale pre-training to perform effectively with limited task-specific data. The majority of medical imaging applications encounter this limitation because rare diseases and specialized centers and expensive expert annotation services limit the amount of available labeled data. Methods that require thousands of training examples will fail in this regime, making WORC an ideal benchmark for evaluating data-efficient approaches such as foundation model transfer learning combined with small-data classifiers.

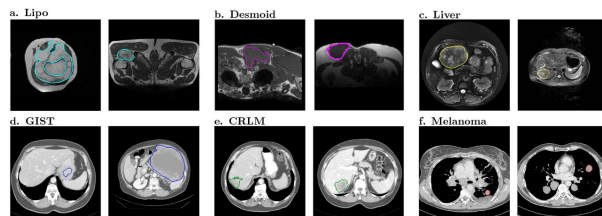


Fig. 1. Example CT and MRI slices from the WORC database showing the diversity of tumor appearances and imaging modalities. Top row: GIST and CRLM (CT). Bottom row: lipomatous tumor and desmoid (MRI).

3 Methods

The pipeline has four stages: data preparation, feature extraction, standardization, and classification, as shown in Figure 3.

3.1 Data Preparation and Preprocessing

The three preprocessing techniques examined to find a good balance between preserving clinical information and improving computational performance are full volume processing, lesion size filtering, and region of interest (ROI) cropping. Full Volume Processing retains the entire scan using a transformation to resize the largest dimension to 128 while retaining the aspect ratio, to preserve

all clinically relevant information but includes tissue that is not related to the lesion which can add noise to the input data. Lesion Size Filtering removes the scans that have small or irregularly-shaped lesions using voxel count and morphologic criteria, which can improve the quality of the signal. Region Of Interest (ROI) Crops extracts fixed sized subvolumes centered at the bounding box of the lesion with adaptive padding [13]. Experiments comparing ROI cropping to other methods determined that ROI cropping is the best option to meet the requirements of SAM-Med3D. By cropping around the tumor in the center of each case, we ensured that the encoder received primarily lesion dominated inputs, which would limit the effect that distant anatomy and scanner specific field-of-view variations could have on the results, and also simplify batch processing and provide consistent contributions from each case. For lesions located near the scan boundary, the ROI margin is asymmetrically clamped to the available image extent on each side, ensuring that only real voxel data is included in the crop. The extracted region is then centered within the target volume and zero-padded to the fixed 128^3 size, so no cases were discarded due to edge proximity.

3.2 Feature Extraction with SAM-Med3D

SAM-Med3D is a foundation model designed specifically for three-dimensional medical image segmentation and feature extraction [17]. The model extends the Segment Anything Model architecture from natural 2D images to volumetric medical data by replacing all 2D operations with their 3D counterparts and adapting the positional encoding to handle three spatial dimensions. The system contains three main components which include a Vision Transformer image encoder that extracts dense embeddings from the input volume and a prompt encoder that processes user-defined guidance through point clicks and bounding boxes and a mask decoder that produces segmentation output. For the purposes of this work, only the image encoder is used to extract feature representations, discarding the prompt encoder and mask decoder components.

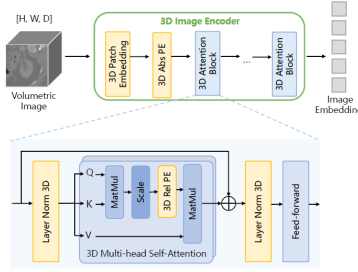


Fig. 2. Architecture of SAM-Med3D. The model consists of a 3D Vision Transformer image encoder, a prompt encoder for interactive guidance, and a mask decoder. For feature extraction, only the image encoder is used.

The image encoder employs a ViT-B backbone, as shown in Figure 2, that divides the input volume into non-overlapping 3D patches, projects each patch into a high-dimensional embedding space, and processes the sequence of patch embeddings through multiple transformer layers with self-attention [4]. The output contains 384 channels which show both local texture details and complete anatomical information at a decreased spatial scale. The dense embeddings need percentile pooling to combine them into single feature vectors which function as compact representations for classification tasks.

The training data comes from SA-Med3D-140K which is a large collection of medical segmentation data that includes 143,000 3D masks which cover 245 anatomical categories and uses CT and MRI imaging modalities [17]. This scale of pretraining far exceeds what is typically available for individual tumor classification tasks and enables the model to learn generalizable representations of medical anatomy and pathology. Second, the model achieves efficient promptable segmentation that requires 10 to 100 times fewer prompt points than previous methods to produce satisfactory 3D segmentation results, indicating that the learned representations capture clinically meaningful structure. Third, the SAM-Med3D-turbo variant used in this work was further fine-tuned on 44 additional datasets to improve cross-domain performance, making it particularly robust to variations in scanner protocols, image quality, and anatomical regions [17].

The choice of SAM-Med3D as the feature extraction backbone for this work is motivated by several considerations that align with the constraints and objectives of the classification task. The primary advantage is the ability to leverage rich pretrained representations without requiring large amounts of labeled tumor classification data. The process of training a volumetric encoder with similar characteristics from start would produce excessive model specialization because the SAM-Med3D encoder starts with a pre-trained model which achieves good results in different body areas and disease types. The model differs from general-purpose vision models because it natively processes 3D volumetric data rather than analyzing individual slices, and was specifically trained on medical imaging rather than natural images. [12, 2].

Another practical consideration is computational efficiency. Pre-trained foundation models, including SAM-Med3D, can be used in a frozen configuration, meaning that feature extraction requires only a single forward pass through the network without gradient computation. This characteristic of foundation models dramatically reduces memory requirements and processing time compared to end-to-end training of a volumetric classification network.

3.3 Feature Standardization and Dimensionality Reduction

The third stage prepares the extracted feature vectors for classification through standardization and dimensionality reduction. Study-level embeddings produced by SAM-Med3D’s encoder have 384 channels and by default are pooled with percentile pooling (capturing five quantiles: 10th, 25th, 50th, 75th, 90th) to yield a 1,920-dimensional vector per case. The vectors undergo standardization to achieve zero mean and unit variance using training partition statistics, which

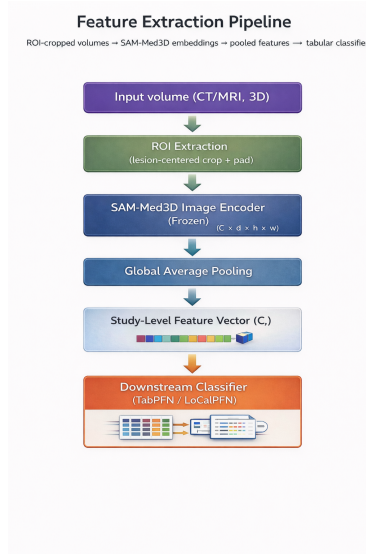


Fig. 3. Feature extraction pipeline using SAM-Med3D. Input volumes are preprocessed with ROI cropping, passed through the frozen image encoder, and aggregated via percentile pooling to produce feature vectors used for downstream classification.

prevent information leakage. Following standardization, PCA reduces the dimensionality while retaining the most informative variance. The standardization process ensures all feature dimensions contribute equally, preventing large numerical values from dominating the subsequent learning process.

The method of principal component analysis serves as an optional procedure which researchers should use when their pooling or aggregation approach generates output data that exceeds typical dimensions (this occurs when researchers combine different lesion embeddings through concatenation or when they replace average pooling with attention schemes or concatenation methods). When triggered, PCA is fitted on the standardized training features and the number of retained components is chosen as the minimum of a preset cap (default 500) and the numerical rank of the training feature matrix [8]. The PCA transform which was trained on the data gets applied to both training and validation features to produce the final reduced representations.

Spatial pooling methods aggregate features within each lesion volume to create a fixed-size embedding. Three aggregation strategies are available—average pooling, multiscale pooling, and percentile pooling. Only average pooling produces a compact enough representation to avoid PCA dimensionality reduction; the other strategies generate higher-dimensional vectors that require PCA before classification. The output of this stage is a set of normalized, optionally reduced feature vectors (one per case), with associated labels and identifiers, which serve as the input to the classification models.

3.4 TabPFN: Transformer-Based Classification for Small Tabular Datasets

TabPFN (Tabular Prior-data Fitted Network) is a transformer-based classifier that uses in-context learning, i.e., during inference, the training set including their respective labels are concatenated with the test feature vector and processed in a single forward pass, resulting in class probabilities without performing gradient-based updates on the parameters [5, 4]. Instead of fitting a model to a given data set, the data set itself is used as a form of context for a pre-trained transformer model that jointly attends to all samples and thus learns implicit decision boundaries in feature space.

The success of TabPFN is due to the design of the training procedure. The model was trained on millions of synthetic classification data sets, each sampled from a well-designed prior distribution called the Prior-data Fitted Network prior, which includes variations in the number of features, class distributions, feature correlation matrices, and non-linear decision boundaries [5]. The training objective of minimizing cross-entropy loss on the held-out samples from each of the synthetic data sets is equivalent to approximating Bayesian posterior inference over the space of possible classifiers. Therefore, this extensive pre-training allows TabPFN to learn how to generalize to new real world data sets without having to train the model on the specific task or adjust the model’s hyperparameters.

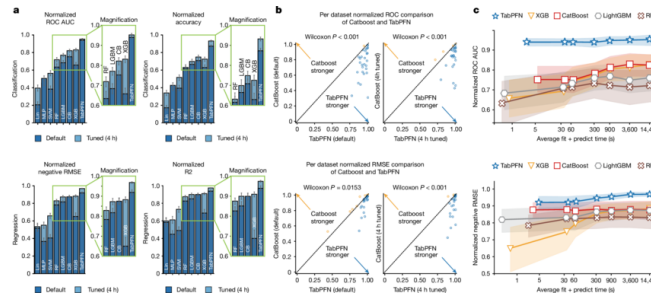


Fig. 4. Performance comparison of TabPFN against traditional machine learning methods. TabPFN achieves competitive accuracy while requiring no hyperparameter tuning and completing inference in significantly less time.

The new approach also pertains to the theoretical underpinnings. TabPFN can be seen as performing an approximate form of Bayesian inference over the posterior predictive distribution of labels based upon the training data [5, 11]. The pre-trained transformer provides a method to approximate the integral of all possible classifiers weighted by their posterior probability given the data (and the prior). It is this statistical basis which describes the success of TabPFN: in contrast to choosing a single hypothesis, TabPFN considers all hypotheses consistent with the data.

TabPFN was selected as the primary classification model for both practical and research reasons aligned with the task of identifying soft-tissue tumors. The advantage of TabPFN may be most pronounced when there are limited datasets. Medical imaging studies on rare tumors typically include a few hundred patients, as opposed to the thousands required for standard deep-learning classifiers to avoid overfitting; TabPFN is specifically designed to operate on datasets containing fewer than 10,000 examples. Therefore, the use of SAM-Med3D embeddings produces the mid-size vectors of features that TabPFN may process after reducing the dimensionality of these vectors, making TabPFN and SAM-Med3D compatible.

The TabPFN integration process occurs after the feature preparation stage in the classification pipeline. After SAM-Med3D embeddings are extracted and pooled to study-level vectors, the features are standardized to zero mean and unit variance using statistics computed on the training set. The analysis uses Principal component analysis to decrease data dimensions into 500 components which fulfills TabPFN feature restrictions and protects against high-dimensional space overfitting with insufficient sample data. The reduced features are passed to TabPFN along with labels, and class probabilities are obtained in a single forward pass.

The model has a hard limit of 10000 training samples, which is not a constraint for the datasets in this study but would require alternative approaches for larger cohorts. Additionally, TabPFN was pretrained on synthetic data with specific distributional assumptions that may not perfectly match all real-world datasets. The thesis evaluates how different classifier selections affect result outcomes through a comparison with LoCalPFN which uses TabPFN to perform local neighborhood retrieval for better results on datasets that show better local than global patterns [16]. Native 3D classification heads trained end-to-end provide an additional baseline that does not rely on the TabPFN architecture.

3.5 LoCalPFN: Local Context Adaptation

Localizing feature space attention can improve performance when there are multiple populations of interest within a feature space, or when different parts of the feature space have different decision boundaries [1]. Medical imaging is one such domain, where tumors can have sub-types, and these sub-types typically reside in completely separate regions of the feature space, often with their own decision boundary [10]. Global approaches do not learn about these local class distribution features and relationships; they rely on global representations of all data points, which can lead to poor accuracy for some classes [16].

TabPFN’s approach to localizing attention to relevant subsets of the data involves two main components: 1) retrieving a neighborhood of data points near the input point of interest; 2) performing inference in TabPFN on this neighborhood [16]. The first component uses K-Nearest Neighbor (KNN) search in the Euclidean space of the standardized features to find the K nearest data points [3], whereas the second component uses TabPFN as described above.

We determine the number of neighbours (K) based on the number of training examples by taking the minimum of 1000 and ten times the square root of the total number of training examples [16]. Based on our dataset sizes (77 to 246 samples per class), this gives approximately 88 to 157 neighbours allowing us to maintain neighbourhood properties while limiting computational overhead [16].

In addition to this we extended the KNN retrieval process with a lightweight adapter which may optionally have been trained on TabPFN logit values [6]. This lightweight adapter modifies the output of the base model based on the knowledge of the neighborhood retrieved, but it does not alter the parameters of the TabPFN model. The adapter was trained for an extremely limited number of iterations using an extremely low learning rate to avoid the possibility of overfitting to local biases in the data [16].

4 Experiments

Six complementary experimental evaluations are performed to test the performance of the SAM-Med3D pipeline for feature extraction against three-dimensional baseline methods, the quality of the feature extraction process using a lightweight classification process, the effects of pre-processing strategies and the types of spatial pooling on the features that were extracted, as well as two probing studies which test the learned representations of the data from the pipeline without a downstream classifier. The same evaluation procedure is used to evaluate all of the experiments including; cross validation over five strata, a 80/20 train/test split with a random seed of 42, and they are tested with three different metrics including; average accuracy, macro F-1, and AUC, across the five folds of cross validation.

4.1 Experiment 1: Transfer-Learning Benchmarks

The main benchmark evaluates four different classification methodologies using matched preprocessing. There are two that follow the pipeline of proposed transfer learning: Med3-TabPFN and Med3-LoCalPFN. These two classification methodologies extract feature vectors from the embedded representations of the frozen SAM-Med3D encoder, and then perform percentile pooling ($5 \times 384 = 1920$ features) and standardization to the training partition. They then reduce dimensionality using PCA (to up to 500 components). Finally, they classify with either TabPFN or LoCalPFN, where LoCalPFN has an additional component that includes k-nearest neighbor context retrieval ($k = \min(1000, 10\sqrt{n_{\text{train}}})$) and an optional lightweight adapter that can be used and is trained on the query data set for 8 epochs on 150 query samples.

There are two other end-to-end classification methodologies in three-dimensional space that also have the same ROI cropped 128^3 volumes as input, but were trained from scratch using cross entropy loss for 200 epochs: 3D DenseNet-121 and 3D ViT.

Each of these four methodologies uses ROI centered cropping with a 30 voxel margin and defaults to a target size of 128^3 . Once features are extracted, the cached embedding is reused to evaluate each of the classifiers individually, allowing the encoder costs to be evaluated independently of the classifier costs.

4.2 Experiment 2: Classification Head on Frozen Encoder

To determine if the SAM-Med3D encoder is capturing discriminative information independently of the tabular classifier, a light-weight fully connected linear head was attached to the frozen encoder output. The head takes the pooled feature vector (percentile pooling, 1920-dimensional) and maps this through one hidden layer with dropout to a binary output.

The training parameters for the experiment are shown in Table 1. Only the parameters of the classification head were trained while the rest of the encoder was frozen. Thus, the only question being asked by this setup is does the simple head itself separate the classes? If the simple head can separate the classes, then there is sufficient discriminative information in the embeddings; otherwise, the bottleneck is upstream.

Table 1. Classification-head training configuration.

Parameter	Value
Encoder	SAM-Med3D-turbo (frozen)
Pooling	Percentile (5 quantiles $\times$ 384 ch.)
Epochs	200
Batch size	4
Learning rate	1×10^{-3}
Weight decay	1×10^{-4}
Dropout	0.3
Optimizer	AdamW
Input size	$128 \times 128 \times 128$
ROI margin	30 voxels

4.3 Experiment 3: Preprocessing Comparison

Three preprocessing strategies are compared to quantify the effect of input preparation on downstream accuracy, all evaluated with TabPFN and percentile pooling:

1. **Full Volume Baseline.** The entire volume was resized to 128^3 . The decoder saw the full field of view which included non-lesion anatomy.
2. **Filtered Full Volume Baseline.** This method was similar to the full volume baseline except cases that did not meet the lesion quality gating criteria

were removed prior to feature extraction. These gating criteria include a minimum number of voxels in the segmentation mask of 300 voxels, a minimum bounding box size of 5 voxels in all dimensions, and a minimum mask density of 0.1.

3. **Region of Interest (ROI)-cropped.** An ROI cropped from around the lesion bounding box, with adaptive padding of 30 voxels, was resized to 128^3 , then z-score normalized. In addition, CT volumes were clipped to $[-1000, 1000]$ HU before normalization.

Once the SAM-Med3D model had been loaded, embeddings for each strategy were sequentially computed to ensure that the model parameters remained the same throughout testing. Accuracy, Macro F1 score, and AUC for all three methods were recorded for each dataset, along with the differences in these metrics for each method versus the full-volume baseline.

4.4 Experiment 4: Pooling Strategy Comparison

Once the ROI cropping was fixed in the pre-processing stage the Spatial Pooling Operator can be analyzed independently. Therefore three different methods are used on the same dense encoder output ($384 \times 8 \times 8 \times 8$ patches):

1. *Average pooling* — Global mean of all spatial locations, which yields a 384 dimensional vector (No PCA necessary).
2. *Multiscale pooling* — Adaptive average pooling at resolution $1 \times 1 \times 1$, $2 \times 2 \times 2$, and $4 \times 4 \times 4$. Then the outputs are concatenated to form a $384 \times (1+8+64) = 28032$ dimensional vector (PCA-reduces prior to classification).
3. *Percentile pooling* — The 10th, 25th, 50th, 75th, and 90th percentile of each channel, produces a $384 \times 5 = 1920$ dimensional vector (PCA-reduces prior to classification).

Classification is performed using TabPFN with 5 fold cross validation with up to 500 PCA components. This will reveal if distributional statistics (percentile), or multi-scale structure (multiscale) yield better results than a basic global summary (average).

4.5 Experiment 5: Attention-Map Analysis

This study examines where the SAM-Med3D encoder assigns attention compared to the lesion mask without developing any classification models. We capture attention weights using a "monkey-patch" technique to modify all self-attention layers in the ViT-B encoder (12 blocks, 12 heads per layer). Due to the fact that the first three (0 1, 3 4, 6 7) and last two (9 10) of the 12 blocks operate as windowed attention with a padded token space of $14^3 = 2744$, we will be rolling up only the four global blocks (2, 5, 8, 11) which process the full $8^3 = 512$ -token sequence.

We compute the rollout by multiplying the global blocks multiplicatively as follows:

$$R = \prod_{l \in \{2,5,8,11\}} \left(\frac{1}{2}A_l + \frac{1}{2}I \right)$$

Here, A_l represents the average attention head matrix at the l -th block and the identity term corresponds to the residual connection. After computing the resultant 512×512 rollout matrix, we perform a column-wise aggregation (mean attention per token) and then reshape it to the 8^3 -patch grid. Finally, we tri-linearly up-sample to the original 128^3 volume to generate a spatial attention map.

For each case, we report two additional measures over all 930 studies:

1. **Attention Mass:** the fraction of normalized attention assigned within the binary lesion mask.
2. **Concentration Ratio:** attention mass divided by lesion volume fraction; values greater than 1 indicate an abnormal amount of attention assigned to the lesion.

4.6 Experiment 6: Embedding Diagnostics

Three complementary statistical tests investigate whether there is any information about label in the 1920 dimensional embeddings, each assessing structural information from a different scale:

1. **Mann—Whitney U (Feature Level).** A two sided Mann—Whitney U Test is performed on each of the 1920 dimensions independently to see if there are differences in the class 0 and class 1 distribution. The p-values are then corrected for multiple comparisons using the Benjamini-Hochberg False Discovery Rate at $\alpha = 0.05$. The effect size is measured as the rank biserial correlation coefficient $r = 1 - 2 * U / [n_0 * n_1]$. This test was conducted on the original data (before PCA), because the rank based statistic used in the Mann—Whitney U test is unaffected by the number of features.
2. **HSIC / CKA (Global Dependence).** The Hilbert-Schmidt Independence Criterion (HSIC) is computed with an RBF kernel (with bandwidth defined by the median heuristic) and a delta kernel on the labels to determine if the full PCA reduced feature matrix (≤ 500 components) and the label vector are statistically dependent. Centered Kernel Alignment (CKA) normalizes HSIC to be in the range $[0, 1]$. The significance of the test is established through a permutation test with 1000 random shuffles of the labels.
3. **k NN Agreement (Local Geometry).** For each sample the k nearest neighbors in standardized PCA space were found using Euclidean distance ($k \in 3, 5, 7, 10, 15, 20$). The proportion of neighbors that had the same label is recorded. An adjusted agreement score (analog of Cohen’s κ) is calculated to account for the imbalance in class prevalence: $\kappa = (\text{observed} - \text{expected}) / (1 - \text{expected})$. Expected = $\exp = \sum_c p_c^2$. The significance of the test was evaluated using a permutation test with 1000 random shuffles of the labels.

Table 2. Experiment 1 classification results on six WORC datasets with ROI-cropped 128^3 volumes (30-voxel margin). All methods report stratified 5-fold CV means (seed = 42); end-to-end baselines are trained for 200 epochs per fold.

Method	Metric	Dataset (n)						Avg
		CRLM (77)	Desmoid (203)	GIST (246)	Lipo (115)	Liver (186)	Melanoma (103)	
Med3-TabPFN	Acc	.508	<u>.694</u>	.597	.504	.597	.476	.563
	AUC	<u>.491</u>	<u>.659</u>	.652	.466	.605	.526	.567
Med3-LoCalPFM	Acc	†	.645	<u>.642</u>	†	.559	†	<u>.615*</u>
	AUC	†	.645	<u>.672</u>	†	.600	†	<u>.639*</u>
3D DenseNet-121	Acc	.821	.813	.760	.783	.694	.690	.760
	AUC	.767	.791	.749	.824	.660	.596	.731
3D ViT	Acc	<u>.531</u>	.680	.598	<u>.557</u>	<u>.608</u>	<u>.582</u>	.593
	AUC	.450	.564	.654	<u>.492</u>	<u>.617</u>	<u>.543</u>	.553

5 Results

Transfer-Learning Benchmarks Table 2 shows the average accuracy of each approach–dataset pair as well as their corresponding average ROC-AUC. The end-to-end approaches clearly outperform the frozen-encoder transfer-learning approaches by a substantial margin.

We can see that 3D DenseNet-121 provides the highest average accuracy as well as average Area Under the Curve (AUC) among all models (76.0% and 0.731, respectively). In addition, we can observe that this model shows the most consistent performance across all of the data sets tested, achieving a minimum accuracy of 69.0% (on the Melanoma dataset) and a maximum accuracy of 82.1% (on the CRLM dataset), and similarly, a minimum AUC of 0.596 (on the Melanoma dataset) and a maximum AUC of 0.824 (on the Lipo dataset).

On the other hand, 3D ViT presents a more consistent but relatively weak overall performance when performing cross-validation with five folds, with a maximum accuracy of 68.0% (on the Desmoid dataset) and a minimum accuracy of 53.1% (on the CRLM dataset), and an average AUC of 0.553. 3D ViT also achieved its highest AUC value on the GIST dataset (AUC = 0.654) and its lowest AUC value on the CRLM dataset (AUC = 0.450, lower than random guessing).

In comparison to the end-to-end methods, the general performance of the transfer-learning based methods was much poorer. On average, Med3-TabPFN provided an accuracy of 56.3% and an average AUC of 0.567, with a best result on the Desmoid dataset (accuracy = 69.4%, AUC = 0.659) and the poorest results on both the Melanoma and CRLM datasets (accuracy = 47.6% and 50.8%, respectively; AUC = 0.526 and 0.491, respectively, both approximately at random level).

Table 3. Experiment 2 results: classification head on frozen SAM-Med3D encoder with ROI-cropped 128^3 inputs (30-voxel margin), trained for 200 epochs. Evaluation uses stratified 5-fold cross-validation (seed = 42). Best result per column in **bold**.

Dataset (n)	Mean Best Ep.	Best AUC	Accuracy
CRLM (77)	6	$.563 \pm .177$	$.415 \pm .086$
Desmoid (203)	77	$.709 \pm .044$	$.665 \pm .064$
GIST (246)	59	$.716 \pm .064$	$.634 \pm .067$
Lipo (115)	63	$.567 \pm .075$	$.478 \pm .106$
Liver (186)	40	$.615 \pm .114$	$.613 \pm .090$
Melanoma (103)	48	$.630 \pm .084$	$.497 \pm .103$
Mean	49	.633	.550

Med3-LoCalPFN was unable to provide any relevant results on the three smaller data sets (CRLM, Lipo and Melanoma), since the number of training samples for each fold was less than the number of neighbors that are needed for retrieval ($k = \min(1000, \lceil 10\sqrt{n_{\text{train}}} \rceil)$) and thus retrieval was not possible. For the remaining three data sets, LoCalPFN improved the AUC over TabPFN by 72 points (from 0.567 to 0.639) but still greatly lagged behind the end-to-end methods.

Therefore, these results show that although the encoder showed a strong tendency to focus on the lesions (see Fig. 5), the pooled and dimensionally reduced features, obtained from the frozen SAM-Med3D model, were insufficiently discriminant for the downstream classification tasks on the WORC data sets. As such, the end-to-end learning strategy, where the task-specific representations are learned through back-propagation and do not depend on pre-trained features, clearly outperforms the transfer-learning based pipeline.

Classification Head Results Table 3 reports classification performance for the frozen-encoder linear head on each of the six WORC datasets, evaluated with 5-fold cross-validation.

Under 5-fold cross validation the frozen-encoder head achieves an average of the best AUC = 0.633 and average Accuracy = 55.0%. Therefore, the frozen SAM-Med3D encoder maintains some discriminative information about tumour classification, at least under conditions of 5-fold cross validation. GIST has achieved the highest best AUC ($\pm 0.064 = 0.716$) and Desmoid has achieved the highest accuracy ($\pm 6.4\% = 66.5\%$), and both have the smallest variation from one fold to another, which indicates that the larger cohorts of GIST ($n = 246$) and Desmoid ($n = 203$) contain the most stable information signal. On the other hand, CRLM is the worst dataset (AUC = $\pm 0.177 = 0.563$) and it has the largest variation between the different folds, including one fold which was down to 0.250 — indicating that the small cohort size of CRLM ($n = 77$) combined with the balanced class distribution, results in a very high sensitivity of the classifier to the particular fold distribution. Lipo is again, as before,

among the more difficult cohorts ($\text{AUC} = \pm 0.075 = 0.567$, Accuracy = 47.8%) and this is consistent with the reported difficulty of distinguishing lipomas from well-differentiated liposarcomas using radiology [9]. Although, the mean best epoch of all the datasets is only 49 of the 200 epochs, the number of epochs required to achieve convergence varies widely among the different folds (for example CRLM requires an average of only 6 epochs to converge while individual Liver and Lipo folds require an average of more than 140 epochs to converge). This large difference in the number of epochs to converge among the different folds suggests that early stopping based on each fold or a learning rate schedule would be preferable to a fixed epoch budget.

Preprocessing Strategy Comparison A comparison of three pre-processing strategies to be used before applying SAM-Med3D Feature Extraction is shown in table 4: (i) The full volume-pipeline using a Baseline strategy; (ii) A Filtered version of the baseline strategy, where masks with less than the minimum voxel count were removed; (iii) An ROI cropping strategy, where a 128^3 voxel-subvolume was cropped, centred at the lesion with a 30 voxel margin. All versions used Percentile Pooling, PCA, and TabPFN, with stratified k-fold cross validation.

Because CRLM has fewer than five cases per class when aggressive mask filtering was applied, it could not have been included in the filtered strategy.

ROI cropping improved the average ROC AUCs by 3.4 percentage points compared to the baseline (0.532 vs. 0.566) and performed better on four of the six datasets (GIST, Melanoma, Desmoid, CRLM).

The largest improvements in the ROC AUC were seen in the smaller lesion datasets, such as GIST (+5.8 pp AUC) and Melanoma (+13.6 pp AUC) where the full volume baseline diluted the lesion signal within a predominately healthy volume.

Only Liver had a lower performance (-7.4 pp AUC) when ROI cropping was applied to it, because its diffuse infiltrative nature benefited from being able to capture a larger anatomical volume.

Although the filtered baseline resulted in an improvement in accuracy on some of the datasets, it consistently lowered the Macro F1 values, suggesting that the class distribution was skewed rather than the features themselves improved.

Thus, ROI cropping with a 30 voxel margin is chosen as the new default pre-processing method to be used in all of the upcoming experiments.

5.1 Pooling comparison

Table 5 demonstrates a comparison of the three evaluated pooling methods (average, multiscale, percentile) with TabPFN applied to ROI cropped image volumes using stratified k-fold cross validation with PCA dimensionality reduction. Average pooling (384-d), through ignoring the distributional aspects, has consistently been outperformed by the other two methods. Multiscale pooling (28 032-d $\rightarrow$

Table 4. Preprocessing comparison (TabPFN, percentile pooling, k -fold CV). Base = full-volume baseline, Filt = lesion-size filtered baseline, ROI = ROI-cropped (128^3 , margin 30). Best result per dataset in **bold**. CRLM filtered is omitted (too few cases after filtering).

Dataset	Accuracy			Macro F1			ROC AUC		
	Base	Filt	ROI	Base	Filt	ROI	Base	Filt	ROI
CRLM	0.481	—	0.508	0.437	—	0.464	0.454	—	0.491
Desmoid	0.680	0.740	0.694	0.502	0.425	0.613	0.645	0.515	0.659
GIST	0.553	0.574	0.617	0.537	0.381	0.617	0.596	0.563	0.654
Lipo	0.504	0.470	0.504	0.497	0.388	0.497	0.481	0.337	0.466
Liver	0.645	0.667	0.597	0.644	0.400	0.592	0.679	0.558	0.605
Melanoma	0.397	0.533	0.476	0.384	0.467	0.462	0.390	0.300	0.526
Mean	0.543	0.597	0.566	0.500	0.412	0.541	0.541	0.455	0.567

Table 5. Pooling strategy comparison (TabPFN, ROI-cropped, k -fold CV). Avg = Average Pooling (384 features), Multi = Multiscale Pooling (28032 features), Perc = Percentile Pooling (1920 features). PCA dimensionality reduction applied before classification.

Dataset	Accuracy			Macro F1			ROC AUC		
	Avg	Multi	Perc	Avg	Multi	Perc	Avg	Multi	Perc
CRLM	0.376	0.651	0.494	0.349	0.617	0.480	0.406	0.641	0.492
Desmoid	0.656	0.714	0.670	0.537	0.639	0.559	0.670	0.693	0.707
GIST	0.557	0.618	0.642	0.555	0.616	0.641	0.595	0.669	0.710
Lipo	0.591	0.565	0.617	0.586	0.557	0.604	0.604	0.608	0.645
Liver	0.581	0.554	0.586	0.570	0.543	0.582	0.577	0.574	0.619
Melanoma	0.610	0.525	0.611	0.600	0.519	0.603	0.655	0.558	0.650
Mean	0.562	0.604	0.603	0.533	0.582	0.578	0.584	0.624	0.637

PCA) and percentile pooling ($1920\text{-d} \rightarrow \text{PCA}$) have demonstrated comparable accuracy and macro-F1; however, percentile pooling has yielded the highest mean ROC AUC (.637 vs .624) at a cost of requiring 15 times fewer raw features. Thus, percentile pooling will be used as the default pooling method for all subsequent experiments.

5.2 Attention Map Analysis

Across all 930 cases in the attention rollout, it can be confirmed that the frozen SAM-Med3D encoder will preferably attend to lesion regions based on the data provided without specific supervision to perform the tasks that were previously completed, with an average concentration ratio of $4.1\times$ (greater than the chance value of 1.0).

The preference or bias for where the encoder prefers to focus varies by the dataset. CRLM showed the greatest amount of concentration ($5.5\times$) with relatively small lesions ($\sim 0.5\%$ of the total volume), suggesting that the encoder is

most effective at identifying compact, well-defined lesions. Lipo and Liver also showed strong concentration ($4.9\times$ and $3.9\times$ respectively), while Desmoid exhibited moderate concentration ($4.7\times$). GIST showed a concentration of $3.9\times$ with larger lesions that would lower the ratio. Melanoma was found to be an exception ($\sim 1.5\times$, close to the random baseline), likely due to its metastasizing pattern on CT scans that is less compact.

Rollout consistently outperformed single-block attention ($4.1\times$ vs. $3.1\times$ for the final block), indicating that the relevant attention for the lesion accumulates as one goes deeper through the layers of the model. The scatter plot (Figure 5) shows that almost every case lies above the diagonal line ($y = x$), indicating that the encoder is allocating more attention than what would be expected based solely on volume. However, it is important to note that attending to the lesion is necessary, but not enough the encoder must also encode *label-discriminative* features within the regions of interest.

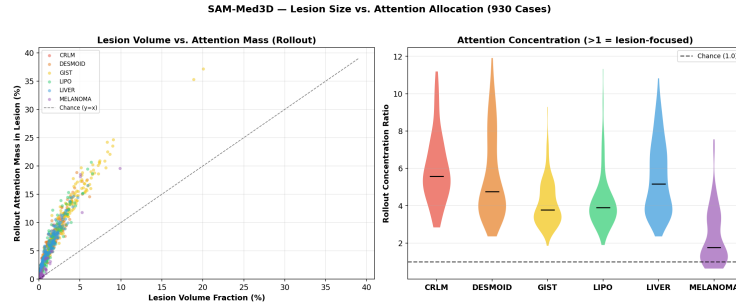


Fig. 5. SAM-Med3D attention allocation relative to lesion size across 930 cases from 6 datasets. *Left:* Scatter plot of lesion volume fraction vs. rollout attention mass within the lesion. Points above the dashed diagonal ($y = x$) indicate attention allocation exceeding chance level. *Right:* Violin plots of the rollout concentration ratio (attention mass / lesion volume fraction) per dataset.

5.3 Embedding Diagnostics

The 1920 dimensional embeddings were tested with three different statistical tests to see if they contained "signal" from the labels used during training. Unfortunately, there was very little discriminative information present in the embeddings that could be used to classify images based on the labels used to train the model. Only two of the six datasets showed some evidence of discriminative information and neither of those had enough of a coherent neighborhood structure.

Mann—Whitney U test. Only Desmoid (395/1920 dims, 21%) and GIST (679/1920 dims, 35%) showed statistically significant separations per dimension using the

Benjamini-Hochberg correction method. The other four datasets showed no statistically significant separations per dimension.

HSIC/CKA. Both the Desmoid (CKA = 0.052, $p = 0.0012$) and GIST (CKA = 0.08, $p = 0.0001$) datasets confirmed that there are weak relationships between the feature and label data for these two datasets. Both CKA values fell in the weak range (0.02 – 0.1). The CRLM dataset had a p-value of 0.083 and thus approached but did not reach statistical significance. None of the other three datasets reached statistical significance ($p > 0.15$).

k-NN Agreement. None of the datasets had strong local label coherence. Desmoid and GIST had negative adjusted agreement at $k = 5$ ($\kappa = -0.037$ and $\kappa = -0.008$) meaning that their signal was not geometrically localized. Lipo is the one exception ($\kappa = 0.113$, $p = 0.021$); it has a weak local pattern that the univariate tests and global kernels do not detect.

Table 6. Embedding diagnostic summary. Three tests probe SAM-Med3D feature quality at complementary levels: Mann-Whitney U (per-dimension), HSIC/CKA (global dependence), and k NN agreement (local geometry). Signal indicators: ● strong, ● weak, ● none.

Dataset	n	MW sig. dims	CKA	HSIC	p	kNN adj. ($k=5$)	kNN p
CRLM	77	0/1920	● 0.069	0.0830	●	−0.046	0.6793 ●
Desmoid	203	395/1920	● 0.052	0.0012	●	−0.037	0.7443 ●
GIST	246	679/1920	● 0.080	0.0001	●	−0.008	0.5255 ●
Lipo	115	0/1920	● 0.027	0.3245	●	0.113	0.0210 ●
Liver	186	0/1920	● 0.023	0.1575	●	−0.082	0.9860 ●
Melanoma	103	0/1920	● 0.033	0.3305	●	−0.037	0.6683 ●

6 Conclusion

SAM-Med3D was evaluated as a frozen volumetric feature extraction method for tumor classification in six WORC benchmark cohorts (930 studies), using a ROI-centric preprocessing pipeline and downstream small-data classifiers. The key finding was that, under the conditions of this investigation, frozen SAM-Med3D representations do not provide enough label-based discrimination to be comparable to end-to-end 3D baseline methods in terms of the clinical subtlety of these tasks, and the limiting factor is upstream (the encoder representation) and not related to the type of downstream classifier.

Experiment 1 showed that the transfer-learning pipeline (the frozen SAM-Med3D embeddings followed by pooling, standardization/PCA, and TabPFN classification) resulted in generally modest classification performance for all of the cohorts studied, and the results were significantly lower than those obtained

by the end-to-end 3D classifiers, which were specifically trained for each task. Although ROI cropping and use of stronger pooling operators improved the results relative to the fully processed volume results, they did not improve the results enough to bring them in line with those obtained via end-to-end learning.

Experiment 2 strengthened this interpretation. When a lightweight classification head is trained on top of the frozen and pooled SAM-Med3D embeddings, the average classification performance is relatively modest (and has significant variation on the smallest cohorts), and the performance is not comparable to the end-to-end baselines. Since this experiment removed the tabular stage (i.e., PCA and TabPFN/LoCalPFN) and asked whether a simple supervised head could reliably distinguish between classes based upon the frozen features, the results support the conclusion that the primary bottleneck is the SAM-Med3D as used here (frozen, segmentation-pretrained, and summarized through fixed pooling), and not simply the downstream modeling choice.

An alternative explanation for this discrepancy in performance is a type of task mismatch: SAM-Med3D has been trained on localizing and segmenting lesions from medical images; it has therefore been optimized to learn features about the shape and contours of lesions that are sufficient to enable the creation of masks but not necessarily the most discriminative signal for identifying subtypes (i.e., the internal texture, heterogeneity or contextual characteristics of the lesion that are often difficult to identify visually). Therefore, if SAM-Med3D learns to represent an image through "where the lesion is" and "what does the border look like", then using fixed pooling over frozen features will likely eliminate or diminish the discriminative fine-grained cues necessary for distinguishing among the relatively clinically subtle labels for subtypes.

The probing results are consistent with this conclusion. The attention rollout demonstrated that the encoder allocates disproportionately high amounts of attention to the lesion region without explicit task-supervision, i.e., the model is not ignoring the tumor. The embedding diagnostics, however, indicate that the attention to the lesion is insufficient: across several cohorts there is weak or inconsistent feature, label dependence and limited local label coherence, indicating that class-relevant cues are either not being robustly encoded or are being encoded in a geometry that is difficult to exploit in small-sample settings.

Therefore, if the research in this direction is continued with additional foundation models from the same family, the focus should be shifted from "frozen features + downstream classifier" to adapting the representation to the classification objective. The most direct next step would be to fine-tune the SAM-Med3D encoder (either fully or partially) on the target tasks, preferably using data-efficient techniques to minimize over-fitting risks on the small cohorts, and to combine this with pooling/aggregation mechanisms that maintain class-relevant spatial structure and avoid reducing it to fixed summary statistics.

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